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Influence of crystalline structure on creep resistance capability in semi-crystalline Polymers: A coarse-grained molecular dynamics study

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ABSTRACT

In this study, we investigated the intrinsic correlation between the crystal configuration and persistent mechanical stability of semi-crystalline polymers. The creep resistances of microstructures with different grain sizes and the same crystallinity were evaluated using coarse-grained molecular dynamics simulations. It was demonstrated that under tensile loading at constant pressure, microstructures with larger grains have more pronounced resistance to molecular rearrangement and significantly delay creep deformation. The improved creep resistance can be attributed to two factors. First, larger crystal sizes result in an increased moment of inertia, reducing the angular velocity required for the rotational rearrangement of the crystalline phase. Second, the elongated crystalline stems enhance the resistance to intermolecular slippage, elevating the strain energy necessary to disrupt the crystalline structure. These findings reveal the molecular basis of creep resistance at the nanoscale and highlight the pivotal role of crystal morphology in enhancing the long-term mechanical integrity of polymers.

1. Introduction

With the increasing use of polymers in long-term, high-load-bearing designs, dimensional stability under static loading has become increasingly important [1,2,3,4]. Thermoplastic semi-crystalline polymers are promising materials owing to their low density, excellent thermal–mechanical properties, and high resilience to corrosive environments [5,6]. They are being actively evaluated for use in various applications, including automotive honeycomb panels, pressurized pipes for oil and gas transportation, and supportive biomedical implants [7,8,9]. From a microstructural perspective, semi-crystalline polymers exhibit a dual-phase composition comprising a rigid and brittle crystalline phase alongside a flexible and ductile amorphous phase. Therefore, a comprehensive understanding of the microstructural response under static loading conditions is pivotal for a robust mechanical design with sustainable performance.

The macroscopic long-term mechanical properties of semi-crystalline polymers are intricately related to interactions within the molecular-scale domain. Thus, significant efforts have been dedicated to understanding the creep behavior of semi-crystalline structures at the microscopic level [10,11,12,13,14]. Free-volume theory has been proposed as the predominant mechanism governing creep deformation in amorphous domains [15]. This model suggests that sustained deformation under static pressure leads to polymer chain disentanglement and the nucleation of nanoscale voids. Subsequent abrupt failures are promoted by void growth and coalescence. Jia et al. [16] experimentally investigated the effect of the crystalline phase on creep deformation. At high strains just prior to abrupt failure, the lamellar stacks inclined toward the macroscopic orientation and experienced significant intermolecular slippage. Intermolecular slip severely impairs the structural stability of semi-crystalline polymers because even low-energy changes lead to large deformations. Pan et al. demonstrated that the creep resistance of polymers could be significantly enhanced by modulating the crystalline phase [17]. The molecular orientation and distinct shish-kebab structure during oscillatory shear injection molding hinder the polymer chains from slipping or pulling against external loads, effectively resisting the dislocation and cleavage of the crystal molecules.

Extant research on semi-crystalline polymers has focused on the precise manipulation of crystalline morphology at the nanoscale level. These observations demonstrated that reduced cooling rate, thermal annealing, shear flow, and pressing during crystallization can increase the effective thickness of isotropically grown lamellae [18,19,20,21].

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Mechanical processing techniques such as drawing or stretching can impart a macroscopic orientation to polymer crystals, elevating the crystallinity within the microstructure to a theoretical limit [22]. The introduction of heterogeneous nucleation agents also modulates the orientation and periodicity of the polymer lamellae [23,24]. The abovementioned attempts show that the microstructure of the crystal phase can be widely controlled through a delicate control of the path of the crystallization process.

To rationally design semi-crystalline polymers with mechanical sustainability, a comprehensive understanding of the effects of crystallinity, crystal shape, size, and distribution state on the creep response is essential. To achieve this, the underlying mechanisms whereby the crystal structure contributes to creep resistance performance must be elucidated. However, a direct correlation between crystal structure and persistent mechanical stability has yet to be established. Microscopic rearrangements, slips, and unwinding of molecular chains cause continuous deformation of the polymer below the yield; however, it is difficult to capture the motion of these individual segments in bulk measurements [25]. Similarly, the significance of crystal morphology design for robust creep resistance is not well understood.

Motivated by the above, in this study, we investigated the influence of crystalline structure on creep resistance performance *in silico*. Using coarse-grained molecular dynamics (MD) simulations, highly crystalline polyethylene (PE) microstructures with various crystal grain sizes and similar crystallinity levels were prepared. The static mechanical properties under a constant strain rate and the creep behavior under a constant tensile pressure were comprehensively evaluated. The findings clearly reveal that the crystalline phase structure can be a predominant factor in creep resistance despite exerting a negligible effect on the stress–strain profiles. Structural analysis revealed that the improved creep performance in the extended crystals was due to the enhanced resistance to rotational motion and suppressed intermolecular dissociation during loading. Consequently, this study provides fundamental insights into the molecular-level role of the crystal structure in the creep deformation of semi-crystalline polymers.

2. Models and methods

2.1. Construction of isotropically crystallized polymer microstructures

A linear PE molecule comprising 1000 monomers was modeled in an MD simulation environment. The PE chains used in this study were prepared in a coarse-grained framework in which each C_2H_4 unit was treated as a single particle with a molar mass of 28.05 g/mol. Subsequently, 96 molecules were randomly positioned in a periodic cubic unit cell. To prevent unintended interference between molecules during initial positioning, the unit cell was set to be sufficiently large with a density of 0.1 g/cc. The molecular behavior of the coarse-grained chains was described using the parameter sets proposed by Moyassari et al. [26,27]. The applied parameters proved to be effective in producing reliable results regarding the crystallization behavior, crystalline morphology, and mechanical properties [26,27].

The potential energy of the entire unit cell was minimized using the conjugated gradient method, with an energy cutoff of 10 kcal/mol. Finally, the unit-cell system was progressively densified in an isothermal–isobaric (NPT) ensemble at 300 K and 1 atm for 20 ns. The motions of the constituent atoms during time integration were calculated using the Verlet algorithm with a time step of 1 fs. Non-isothermal crystallization simulations were performed to transform the equilibrated polymer structure into an isotropically crystalline polymer structure. The equilibrated PE model was exposed to a high temperature of 450 K for 25 ns in the NPT ensemble. Because the temperature sufficiently exceeded the melting point of typical high-density PEs (399–406 K, [28,29]), the chains within the unit cell became completely amorphous. The model was then gradually cooled at cooling rates of 1, 10, and 100 K/ns. After the temperature reached 300 K, it stabilized until the total simulation time reached 175 ns.

Other details for performing MD simulations are as follows: The commercial MD software Materials Studio 2020 (Dassault Systèmes, France) was used for initial modeling of the microstructure. The open-source MD code LAMMPS (Sandia Labs, USA) was used to simulate the crystallization and perform static or long-term mechanical testing of the polymer [30]. The molecular orientation and related properties of the crystalline polymers were visualized using the OVITO software package [31].

2.2. Evaluation of molecular order in modeled crystalline structures

The evolution of the polymer backbone into a regularly arranged structure was evaluated using the Steinhardt bond-order parameter ($q_6q_6^*$) as proposed by Zhang and Larson [32]. This parameter is a correlation function that determines the value of a reference atom based on the spatial configuration of the neighboring atoms. The detailed process for measuring $q_6q_6^*$ is as follows: First, the q_{6m} complex vector for atom i constituting the polymer chain was defined using the following equation:

$$q_{6m}(i) = \frac{1}{Ni}\sum_{j=1}^{Ni} Y_6^m\left(\phi_{i,j}, \varphi_{i,j}\right) \tag{1}$$

where Ni represents the number of neighboring atoms within a distance cutoff of 0.54 nm from the atom i, Y_6^m denotes the m component of the six-degree spherical harmonic, and $(\phi_{(i,j)}, \varphi_{(i,j)})$ is a spherical polar orientation of a vector connecting the atoms i and j. Then, the bond-order parameter $q_6q_6^*$ of atom i was calculated as follows.

$$q_6q_6^*(i) = \sum_{j=1}^{Ni}\sum_{m=-6}^{6} q_{6m}(i)q_{6m}^*(j) \tag{2}$$

where the asterisk (*) indicates complex conjugation. $q_6q_6^*$ approaches 0 for the amorphous phase close to the liquid phase, whereas it assumes a positive value for the crystalline constituents. In other words, it can quantitatively describe the transition of polymer chains from amorphous to crystalline structures in nanoscale domains.

3. Results and discussion

3.1. Crystallinity analysis of modeled PE unit cells with different cooling rates

The cooling rate applied during melt crystallization significantly influenced the microscopic structure of the crystalline phase. Typically, faster cooling rates lead to more nuclei seeds, smaller, denser crystals, and reduced lamellar thickness in crystalline polymer systems. To investigate the influence of the crystal structure on the mechanical creep behavior, PE microstructures were prepared at cooling rates of 1, 10, and 100 K/ns (Fig. 1**(a)** top). The conversion of amorphous polymer chains into regularly ordered microstructures during cooling was quantitatively characterized using the bond-order parameter $q_6q_6^*$.

During crystallization at a constant cooling rate, the distribution of $q_6q_6^*$ at each temperature step adequately represented the state of the microstructure (Fig. 1**(a)** bottom). The PE polymer completely melted in the initial high-temperature environment (450 K) and was therefore considered to be in a completely amorphous arrangement. This was also confirmed by the simple unimodal distribution, where the $q_6q_6^*$ profile was centered at a value of 0. However, as the temperature decreased, regularly ordered microstructures gradually grew as the intermolecular interaction energy became more dominant than the given thermal energy. In particular, the peak near 0 showed a sharp decrease at 400 K, which is the crystallization point of typical high-density PE [28,29]. The shape of the $q_6q_6^*$ profile was converted into a bimodal distribution, which also supported the appearance of significant crystalline phases.

The thermal vibrations of the constituent atoms caused large perturbations in the calculation results of $q_6q_6^*$ values. To mitigate thermal

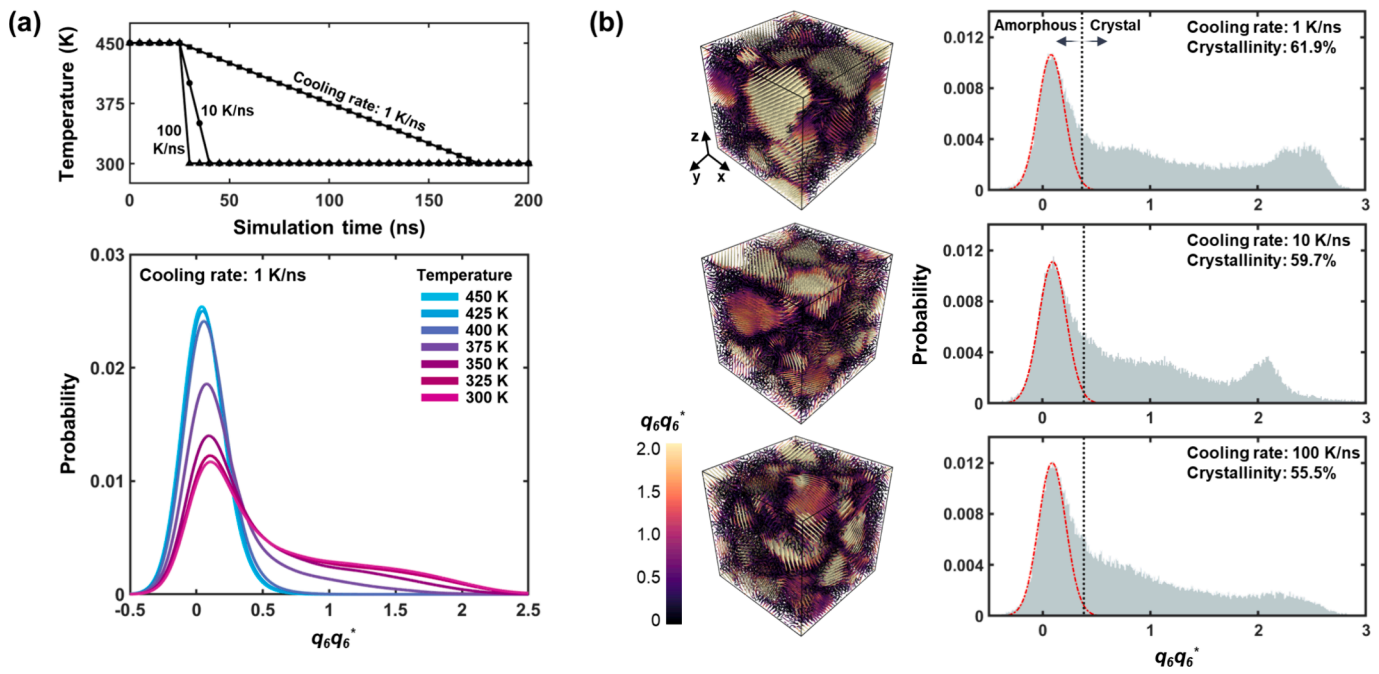


Fig. 1. Morphological analysis of semi-crystalline PE during non-isothermal crystallization. (a) Temperature profiles over simulation time at various cooling rates (top) and changes in $q_6q_6^*$ distribution during cooling at a constant rate of 1 K/ns (bottom). Each obtained $q_6q_6^*$ profile was interpolated with kernel density estimates. (b) Molecular configuration of the PE models (left) and the corresponding distribution of $q_6q_6^*$ (right) after crystallization was complete. The vertical dotted lines in the $q_6q_6^*$ profiles represent the cutoff to distinguish between amorphous and crystalline phases, and the values are 0.367, 0.382, and 0.380 in that order from the top.

effects and focus only on the structural features of the semi-crystalline structure, each crystallized PE model was rapidly quenched under the NVT ensemble for 5 ns at 0.1 K. The calculated $q_6q_6^*$ distributions after equilibration under cryogenic conditions are presented in Fig. 1**(b)**. For all the conditions considered, two distinct peaks and an intermediate transient region were observed in the $q_6q_6^*$ distribution. In particular, as the cooling rate increased, the right-hand peak value decreased and became less distinct, indicating that the crystal structure was more fragmented and poorly assembled (see **Figure S1** for the number and average length of the crystals). To distinguish between the amorphous and crystalline phases based on the results, we fit the lower peak near zero using a Gaussian function. All areas except the 99 % confidence interval of the Gaussian curve were considered crystalline regions. The crystallinity was calculated as the ratio of crystalline atoms to the total number of atoms; the values were 61.9 %, 59.7 %, and 55.5 % at cooling rates of 1, 10, and 100 K/ns, respectively (see **Table S1** for the number of constituent atoms of each phase). This is in good agreement with the experimentally estimated crystallinity of the melt-crystallized high-density PE (50 %–65 %). [33,34]). Furthermore, a comprehensive verification of the implemented semi-crystalline models is provided in **Table S2**.

3.2. Yield stress prediction in quasi-static conditions

The main point of the previous section is that PE microstructures with diverse structural properties but similar crystallinity levels (55.5 %–61.9 %) have been implemented from nanoscale models. In other words, from the prepared models, we can independently explore the creep behavior influenced purely by the crystalline morphology, excluding crystallinity level effects. The next hurdle in analyzing creep tests from models is the accurate derivation of the yield stress of the prepared system. Mechanical deformation tests implemented in MD simulations have very high strain rates compared with experiments, and the resulting yield point is inevitably overestimated. Therefore, a post-processing strategy that can predict the quasi-static properties of a system based on MD simulation results is essential.

We used a method that combined the viscoelastic model with the change in the yield point according to various strain rate conditions. First, uniaxial tensile tests were performed on PE models stabilized at room temperature. Under an NPT ensemble, tensile deformation with a constant engineering strain rate was applied to the unit cell until the maximum strain reached 0.2. While the length of one axis of the unit cell increased at a constant rate, atmospheric pressure was applied to the other two axes to account for the Poisson effect. The angle of the unit cell edge was consistent at 90° throughout all tests. The stress tensors evolved in the constituent atoms were calculated based on the virial theorem, and five different strain rates ranging from 5×10^8 to 10^{10} /s were considered. The elastic modulus was derived by linearly fitting the profile in the low-strain range of 0–0.02. The yield stress at each strain rate was estimated using the 2.5 % offset method, which is widely applied to glassy polymer systems. The stress–strain profile increased monotonically in the elastic region and then converged as it experienced yielding (Fig. 2**(a)**). This behavior was qualitatively consistent regardless of the cooling rate. The similarity in the mechanical responses demonstrated that the crystalline phase contents of the considered models were similar. Also, the elastic modulus of models was 636.8 ± 21.4 MPa, which is very similar to experimental values (609–667 MPa, [35,36]) of typical high-density PE. However, the yield stress of the PE models (28.5 ± 0.4 MPa) was significantly overestimated compared with the actual one (19.7–23.6 MPa, [35,36]). The lower the strain rate, the longer the time required for the microstructure to relax, resulting in a lower yield stress (Fig. 2**(b)**). However, in the region covered by the simulations, it was still significantly higher than the experimentally observed values.

Therefore, to bridge the gap between MD and experimental timescales, the quasi-static yield stress was estimated using three viscoelastic theories: Argon, Eyring, and cooperative models. **Tables S3–S5** summarize the formulae and derivation coefficients for each yield-strength model. The models considered were developed based on the hypothesis that viscoelastic material behavior arises from microscopic molecular

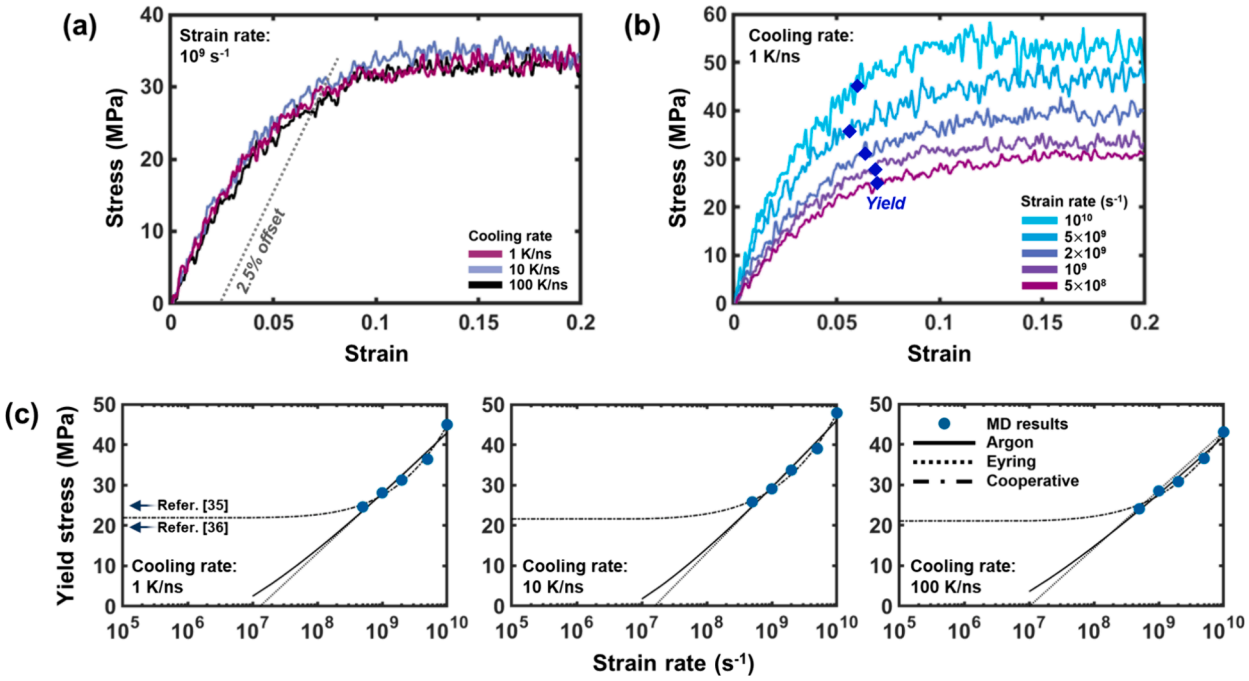


Fig. 2. Mechanical response of semi-crystalline PE under uniaxial tensile simulation. (a) Stress–strain curves of the microstructures prepared under different cooling rates. (b) Strain rate dependence of the mechanical properties of semi-crystalline PE. The yield point in each case is highlighted with a blue dot. (c) Yield stress under quasi-static conditions derived using a combination of viscoelastic model and simulation results.

rearrangements. In particular, these models have demonstrated their effectiveness for polymers in the glassy state, including epoxy, polycarbonate, and poly(methyl methacrylate) [37,38,39].

The yield stress under quasi-static conditions was estimated based on the value at each strain rate (Fig. 2**(c)**). Among the yield models, only the cooperative model produced reliable results. The Eyring and Argon models were built on the assumption that the viscoelastic behavior is strictly proportional to the strain rate and were, therefore, unable to bridge the significant differences between the strain rates covered by our simulations and experiments. Additionally, within a given strain-rate range, PE polymers exhibit nonlinear changes in their mechanical properties. Extrapolation using the cooperative model not only shows such nonlinearity well but also shows high convergence for strain conditions similar to the experimental conditions, making it suitable for predicting the yield strength. The estimated yield stress at cooling rates of 1, 10, and 100 K/ns were 21.9, 21.5, and 21.0 MPa, respectively, comparable to the experimental values of 19.7–23.6 MPa for typical high-density PE [35,36].

3.3. Tensile creep tests for semi-crystalline PE

Tensile creep test simulations were performed to assess the long-term mechanical stability as a function of the crystal configuration. A constant tensile pressure was applied along one axis of the unit cell, and atmospheric pressure was maintained along the other two axes to account for the Poisson effect. The pressure for creep deformation was set to 17.5 MPa, which corresponds to 80 % of the quasi-static yield stress of the systems. The applied loading conditions exclude mechanical failure owing to the static resistance limit, thus making it possible to cover only the subcritical region where failure occurs owing to viscoelastic behavior.

Local deformations within the unit cell during creep were characterized based on the Lagrangian strain tensor. The strain tensor of the reference bead was scalarized to the von Mises strain (ε_{vM}). The neighboring atoms involved in the atomic strain calculations were considered up to a radius of 12 Å, corresponding to the van der Waals cutoff distance.

Under an external pressure below the yield stress, the PE models exhibited continuous elongation along the loading direction over time (Fig. 3**(a)**). During this process, the creep strain evolution exhibited three distinct stages (Fig. 3**(b)**). I) The creep strain of the PE microstructure increased rapidly, then gradually decreased over time, and eventually plateaued. II) The models entered a steady creep stage, where the strain rate remained constant. III) As the creep strain exceeded a critical point, deformation accelerated and led to failure. This three-stage configuration aligns with the typical creep behavior of polymeric materials [40,41] and is consistently observed, irrespective of the cooling rate and load direction.

Interestingly, a significant difference existed in the creep evolution rate depending on the crystal composition of the PE microstructures. Models fabricated at higher cooling rates (i.e., those exhibiting more segmented crystals and shorter constituent stems) significantly accelerated the progression to creep failure. This is also evident from the fact that the slope of the stabilized region observed in stage II becomes steeper at faster cooling rates. To quantify the creep failure threshold, we defined the critical strain as the point at which the creep rate reached 0.65 and rapid failure began (Fig. 3**(c)** left). The results suggest that the critical limit increased as the cooling rate decreased, resulting in a higher resistance to structural failure at a constant load (Fig. 3**(c)** right).

It has been demonstrated that in semi-crystalline systems, the rate of creep evolution is suppressed as the crystals become larger and better organized. In other words, the morphology of the crystalline phase was identified as an important structural factor determining the creep resistance capability. This is in contrast to the fact that, as long as a similar level of crystallinity is achieved, the static mechanical properties are consistent regardless of the size of each crystalline phase. Accordingly, an in-depth understanding of the creep process of semi-crystalline polymers is required to evaluate the influence of the crystalline structure on the long-term behavior of materials.

3.4. Molecular origin of creep resistance mechanism in semi-crystalline polymer

The elongation process of the PE model under constant pressure was

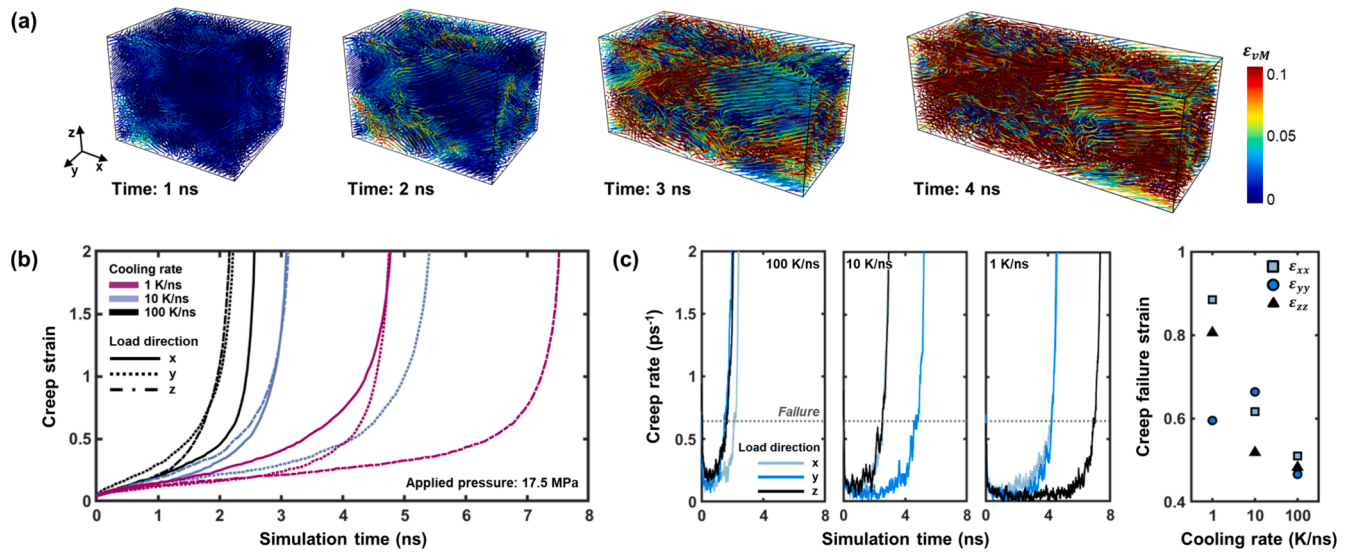


Fig. 3. Analysis of creep behavior exhibited by semi-crystalline PE models. (a) Molecular configuration change under constant external pressure. The presented case is a microstructure prepared at a cooling rate of 1 K/ns, and each bead is colored according to the magnitude of the von Mises strain. (b) Comparison of creep strain evolution of PE models prepared at different cooling rates. (c) Creep rate change with time when exposed to tensile load (left) and associated critical strain (right).

further analyzed by examining the main parameters such as mass density, Harman's orientational order, strain energy density, and non-covalent energy within the crystalline phase. Harman's orientational order (S), which represents the molecular alignment of the polymer backbone chains along the loading direction, was determined by considering the directional vector penetrating the PE beads two places away from the reference bead ($\vec{V}_{i-2,i+2}$). The relative angle (θ) between the directional vector and the pressure direction was measured and quantified as a representative scalar value using the following equation [42,43]:

$$S = \langle \frac{3cos^2\theta - 1}{2} \rangle \tag{3}$$

where the bracket symbol $< >$ indicates the average value. The structural damage accumulation in each polymer subdomain (i.e., amorphous and crystalline phases) during creep deformation was estimated using the strain energy density (u), as the following equation [44,45,46]:

$$u = \int_0^{\varepsilon_{vM}} \sigma_{vM} d\varepsilon \tag{4}$$

where σ_{vM} denotes the von Mises stress of constituent atoms based on virial theory, and ε_{vM} represents the von Mises strain derived from the Lagrangian strain tensor. The amorphous and crystalline phases were clearly distinguished based on the $q_6q_6^*$ bond-order parameter, as previously defined (Fig. 1).

Under constant pressure, the semi-crystalline PE exhibited three notable morphological features (Fig. 4**(a)**). First, the mass density of the system remained nearly invariant throughout the creep process. The strain increment observed during creep testing arose from internal rearrangements aimed at maintaining a minimal energy state. The free volume that emerged as the microstructure underwent stretching was promptly reoccupied by molecular flow. This inhibition of free-void formation was distinct from that observed in the static mechanical

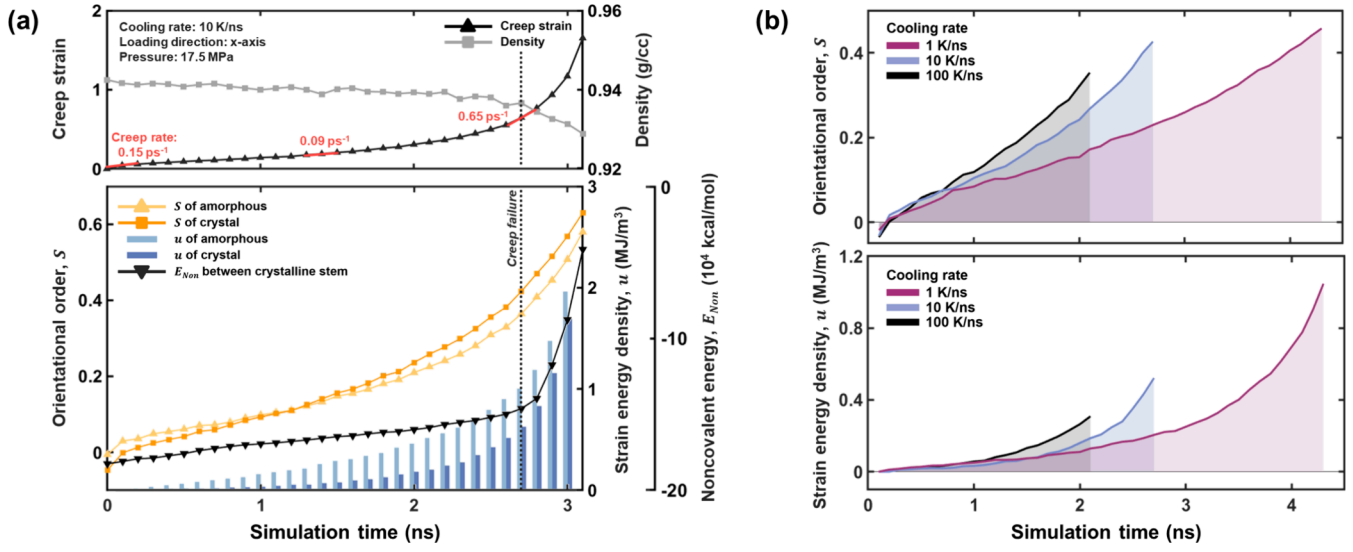


Fig. 4. Structural analysis of semi-crystalline PE according to creep progression. (a) Changes in mass density, Harman's orientational order, strain energy density, and non-covalent energy of the crystalline phase. (b) Molecular alignment (top) and deformation energy accumulation (bottom) profiles of the crystalline segments until creep failure.

tests, where the material stretched monotonically [47,48]. In other words, creep failure in semi-crystalline polymers is independent of the growth and coalescence of nanosized voids, which are the primary factors in uniaxial tensile failure. Second, the orientation order, representing the microscopic alignment of the constituent chains, progressively increased with the creep strain. Interestingly, the orientation magnitude at each creep strain level was approximately the same in both the amorphous and crystalline phases. This suggests that a constant load applied to a semi-crystalline system is distributed uniformly across each phase, promoting microstructural reorientation. Third, the evolution of the structural damage correlates with the extent of creep progression. In the initial stage, the applied load predominantly accumulates as strain energy in the amorphous region owing to the substantial differences in mechanical stiffness. This is evident from the observation that the strain energy of the crystalline phase is negligible during the simulation time of 0–2 ns. However, as the system approached the failure strain, the strain energy of the crystalline phase became more pronounced. This indicates that creep failure is related to the structural destruction of the crystal phase. The importance of the crystal phase in creep failure was also confirmed by the rapid increase in the non-covalent energy of the crystalline chains near the failure point.

The point of the above is that the microstructure of the crystal phase is a critical factor in determining creep resistance ability. To further elucidate this, the changes in the molecular orientation and strain energy accumulation corresponding to the crystal structure were quantitatively analyzed (Fig. 4**(b)**). The results show that smaller and more fragmented grains (i.e., microstructures that experience faster cooling rates) lead to faster molecular orientation under equivalent stress. In other words, smaller crystal clusters have less resistance to rearrangement through rotational motion. Additionally, as the orientation of the crystalline molecules increases, the intermolecular engagement becomes dominated by weak van der Waals interactions. Therefore, the strain energy density results show that the deterioration of the crystal phase intensifies as creep progresses. Longer crystal chains provide enhanced interactions with neighboring atoms, which increase the resistance to inter-molecular slippage. Therefore, the maximum strain energy at which creep failure occurs increases as the cooling rate of the microstructure decreases.

Based on the above findings, the creep failure procedure of semi-crystalline polymers can be summarized as follows (Fig. 5). A constant load applied to the semi-crystalline system was distributed uniformly across each subdomain, promoting microstructural rearrangement. Because of the difference in stiffness, most of the strain was concentrated in the amorphous region, preserving the crystal structure without damage. However, during stretching, the crystal domains gradually rotated along the loading direction and rearranged into a conformation that was vulnerable to intermolecular shear. Owing to the continuous deformation, the stems that make up the polymer crystals slipped, causing rapid creep failure after a certain time. The mechanism of this procedure is very similar to that of the ductile fracture of PE observed experimentally in the low-stress regime.

The identified creep resistance model is distinct from the typical creep behavior for pure amorphous and crystalline structures. (1) The formation and coalescence of free voids are considered key factors governing the creep behavior of amorphous polymers [15]. However, in semi-crystalline polymers, the homogeneously distributed crystalline domains act as spatial constraints for the rearrangement of amorphous segments. Therefore, disentanglement and the associated free void formation are greatly suppressed in semi-crystalline systems. (2) Crystalline chain slippage, which is the main cause of creep deformation in single-polymer crystals [49], is equally observed in current semi-crystalline systems. However, our results suggest that slippage of the crystalline chains is important only in the last stage of creep. The gradual structural evolution during the steady creep stage (i.e., rotation and alignment of the polymer crystals) decreases the stability of the crystalline domain without imposing structural damage. Therefore, understanding the sequential responses based on the stiffness difference between the amorphous and crystalline phases is important in semi-crystalline polymers.

A better-organized crystal structure is advantageous over creep deformation for two reasons. First, the enlargement of the crystal size results in a proportional increase in the moment of inertia for rotational

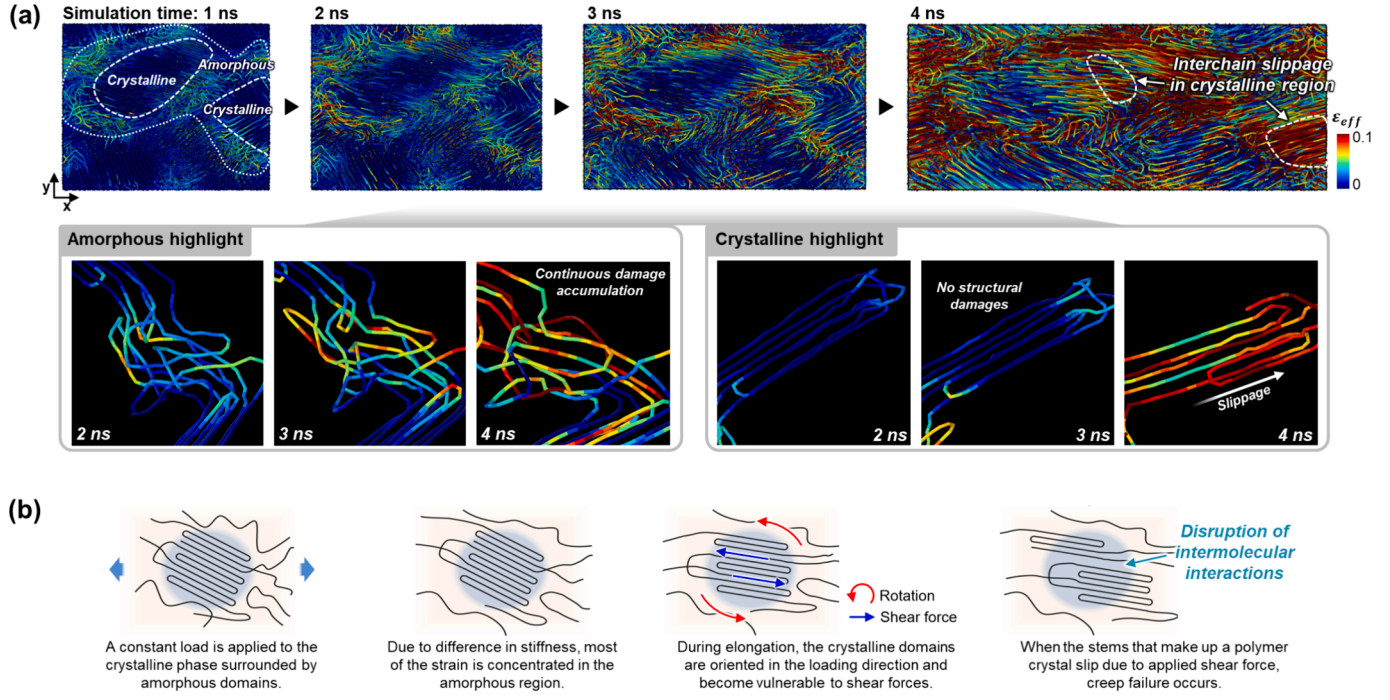


Fig. 5. Creep resistance mechanism of semi-crystalline polymer. (a) Atomistic configuration of the semi-crystalline PE model as creep deformation progression (top, colored by the magnitude of von Mises strain), and the highlight snapshots of the amorphous and crystalline segments (bottom). (b) Schematic of the creep deformation process.

behavior. Therefore, the angular velocity of the crystalline phase decreases under constant pressure, impeding the progression of the steady creep stage. Second, the extended stem length enhances resistance against intermolecular slippage. The strain energy required to disrupt the interaction between the crystalline chains increases, thereby significantly enhancing the maximum strain for creep failure. From an engineering perspective, strategies to increase the lamella thickness, such as thermal annealing, pressing during crystallization, and fibrillation through drawing, can be effective approaches to ensuring the stability of semi-crystalline polymers against creep [18,20,22].

The operating environments for semi-crystalline PE, including gas or oil transport pipes, are characterized by significant creep damage, as opposed to low fatigue damage. Consequently, the gradual evolution of the polymer microstructure under static loading conditions is a critical factor in determining the fatigue life of products. This study highlights the crucial role of the crystalline phase by elucidating the molecular origins of creep at the nanoscale. Therefore, it can serve as theoretical guidance for the design of structures and materials with superior mechanical sustainability and creep-fatigue resistance.

4. Conclusion

This study explores the inherent relationship between the crystal configuration and enduring mechanical stability of semi-crystalline polymers. The importance of crystal morphology design for creep resistance capability was emphasized via molecular-level structural analysis.

Within the framework of the coarse-grained MD simulations, several PE microstructures with similar crystallinity levels were considered, and a comparative analysis was conducted according to the grain size. The results suggest that creep progression was strongly influenced by the size of the crystallites. The well-organized microstructure consisting of large grains was highly resistant to crystal phase rearrangement, resulting in a significant delay in creep deformation. There are two reasons for the positive effect of a well-organized crystalline microstructure on creep resistance. First, enlargement of the crystal size results in a proportional increase in the moment of inertia. The angular velocity for the rotational rearrangement of the crystalline phase diminishes with crystal size under constant pressure, impeding the progression of the steady creep stage. Second, the extended length of the stems of the polymeric chains increases the resistance against intermolecular slippage. The strain energy required to disrupt the interaction between the crystalline chains increases, thereby increasing the maximum strain at which abrupt failure occurs. Therefore, strategies to increase the lamella thickness, such as thermal annealing, pressing during crystallization, slower cooling rates, and fibrillation through drawing, can be an effective way to ensure the long-term stability of semi-crystalline polymers.

To the best of our knowledge, this study is the first attempt to establish a connection between molecular-level configuration and mechanical persistence of polymers. In this regard, this study can serve as theoretical guidance for the rational design of semi-crystalline materials for engineering applications, including lightweight and durable plastic components, reliable structural materials under sustained loads, and mechanical parts that ensure long-term dimensional stability.

CRediT authorship contribution statement

Hongdeok Kim: Writing – original draft, Visualization, Software, Methodology. **Joonmyung Choi:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijfatigue.2024.108517.

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